Interim Analysis

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Motivation: Peaking at the data?

Observe data $X_1, ..., X_n$ over n months, with single X_i collected each month. Assume that

$$X_1, ..., X_n \overset{iid}{\sim} N(\mu, 1)$$

Want to test $H_0 : \mu = 0$ vs. $H_A : \mu > 0$. Initial plan: reject if

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i > z_\alpha$$

Question: Can we look at the data early?

- Let $Z_k = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} X_i$ (info from first k time points)
- If $Z_k > c_1$: reject H_0 and <u>stop</u> (don't collect remaining data)
- If $Z_k < c_1$: collect remaining data, then check $Z_n > c_2$ at end of study

<u>So</u>: $P(\text{reject } H_0) = P(Z_k > c_1 \text{ or } Z_n > c_2)$

Motivation: Peaking at the data?

$$X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, 1)$$

$$Z_k = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} X_i \qquad Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i$$

Decision rule: Reject H_0 if $Z_k > c_1$ or $Z_n > c_2$.

Question: How would I choose c_1, c_2 to control type I error at level α?

<u>Bonferroni</u>: $P(Z_k > c_1 \text{ or } Z_n > c_2) \leq P(Z_k > c_1) + P(Z_n > c_2)$

$\Rightarrow$ choose c_1, c_2 st $P(Z_k > c_1) = P(Z_n > c_2) = \frac{\alpha}{2}$

e.g. if $\alpha = 0.05 \Rightarrow c_1, c_2 = 1.96$

<u>But</u>: Bonferroni is conservative (union bound)

Can we do better?